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United States Patent Application Publication

20250282156

Kind Code

A1

Publication Date

September 11, 2025

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SHEET CONVEYANCE APPARATUS AND IMAGE FORMING SYSTEM

Abstract

A sheet conveyance apparatus includes a conveyance portion, a switching portion, a multi-feeding detection portion, and a control portion. The multi-feeding detection portion disposed upstream of the switching portion in a sheet conveyance direction and detecting multi-feeding in which a first sheet and a second sheet that follows the first sheet overlap with each other. The control portion controls the switching portion for guiding the first sheet and the second sheet to the third conveyance path by switching the switching portion to the second position when multi-feeding is detected by the multi-feeding detection portion. When multi-feeding is detected by the multi-feeding detection portion, the control portion calculates a first time at which a trailing edge of the second sheet in the sheet conveyance direction has been passed through the switching portion, and switches the switching portion from the second position to the first position at the first time.

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Family ID: 96948396

Appl. No.: 19/068283

Filed: March 03, 2025

Foreign Application Priority Data

JP 2024-035939

Mar. 08, 2024

Publication Classification

Int. Cl.: B41J11/00 (20060101); B65H7/12 (20060101)

U.S. Cl.:

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to a sheet conveyance apparatus that conveys sheets, and an image forming system.

Description of the Related Art

[0002] For feeding sheets from a component such as a cassette or a tray, a separation pad or a separation roller is disposed for separating a sheet from others, one by one. However, if a following sheet sticks to a preceding sheet, multi-feeding may occur. In the multi-feeding, the preceding sheet is conveyed in a state where the following sheet overlaps with the preceding sheet. The multi-feeding may cause a defect in image formation, a jam, a failure, and the like. For this reason, Japanese Patent Application Publication No. H11-24506 proposes a configuration in which the conveyance is stopped if the multi-feeding of sheets (paper sheets) is detected.

[0003] However, the configuration proposed in Japanese Patent Application Publication No. H11-24506 and configured to stop the conveyance may reduce the productivity because the downtime occurs. Japanese Patent Application Publication No. 2002-333797 proposes a configuration in which if the multi-feeding is detected, the multi-fed sheets are discharged without performing the formation of images, and the images are formed on the following sheets. Japanese Patent Application Publication No. 2002-333797 also proposes a technique in which the multi-fed sheets are discharged to a predetermined discharging tray other than a specified sheet-discharging tray.

[0004] By the way, for switching the tray from the specified sheet-discharging tray to the other discharging tray and discharging the multi-fed sheets to the other discharging tray as proposed in Japanese Patent Application Publication No. 2002-333797, the conveyance path for conveying sheets is switched by a switching portion. In addition, if a following sheet that is not multi-fed is conveyed, it is necessary to return the switching portion to the original position. However, the timing at which the trailing edge of the multi-fed sheet (e.g., the second sheet of the multi-fed two sheets) varies, depending on the amount of overlap (i.e., the amount of shift) between the multi-fed two sheets. In particular, if the switching portion is returned to the original position before the trailing edge of the multi-fed sheet (e.g., the second sheet of the multi-fed two sheets) passes through the switching portion, the second sheet may be caught between the switching portion and a guide, and stopped. In this case, the following sheet may abut against the second sheet and cause a jam.

[0005] An object of the present invention is to provide a sheet conveyance apparatus and an image forming system that can prevent the occurrence of jam while guiding the multi-fed first and second sheets to a third conveyance path.

SUMMARY OF THE INVENTION

[0006] According to a first aspect of the present invention, a sheet conveyance apparatus includes a conveyance portion configured to convey a sheet, a switching portion configured to be switched between a first position and a second position, the first position being a position at which the switching portion guides the sheet conveyed in a first conveyance path by the conveyance portion, to a second conveyance path, the second position being a position at which the switching portion guides the sheet conveyed in the first conveyance path by the conveyance portion, to a third conveyance path, a multi-feeding detection portion disposed upstream of the switching portion in a sheet conveyance direction and configured to detect multi-feeding in which a first sheet and a second sheet that follows the first sheet overlap with each other, and a control portion configured to

control the switching portion for guiding the first sheet and the second sheet to the third conveyance path by switching the switching portion to the second position in a case where multi-feeding is detected by the multi-feeding detection portion. In the case where multi-feeding is detected by the multi-feeding detection portion, the control portion is configured to calculate a first time at which a trailing edge of the second sheet in the sheet conveyance direction has been passed through the switching portion, and switch the switching portion from the second position to the first position at the first time.

[0007] According to a second aspect of the present invention, a sheet conveyance apparatus includes a conveyance portion configured to convey a sheet, a switching portion configured to be switched between a first position and a second position, the first position being a position at which the switching portion guides the sheet conveyed in a first conveyance path by the conveyance portion, to a second conveyance path, the second position being a position at which the switching portion guides the sheet conveyed in the first conveyance path by the conveyance portion, to a third conveyance path, a multi-feeding detection portion disposed upstream of the switching portion in a sheet conveyance direction and configured to detect multi-feeding in which a first sheet and a second sheet that follows the first sheet overlap with each other, a trailing-edge detection portion disposed upstream of the switching portion in the sheet conveyance direction and configured to detect a trailing edge of the second sheet in the sheet conveyance direction, and a control portion configured to control the switching portion for guiding the first sheet and the second sheet to the third conveyance path by switching the switching portion to the second position in a case where multi-feeding is detected by the multi-feeding detection portion. In the case where multi-feeding is detected by the multi-feeding detection portion, the control portion is configured to switch the switching portion from the second position to the first position based on detecting the trailing edge of the second sheet by the trailing-edge detection portion.

[0008] According to a third aspect of the present invention, an image forming system includes the sheet conveyance apparatus, and an image forming apparatus configured to form an image on the sheet conveyed from the sheet conveyance apparatus.

[0009] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1A is a block diagram illustrating a schematic configuration of an image forming system of a first embodiment.

[0011] FIG. 1B is a schematic cross-sectional view illustrating the schematic configuration of the image forming system of the first embodiment.

[0012] FIG. 2A is a schematic cross-sectional view illustrating positions of sheets in a case where a single-side printing job is performed in the image forming system of the first embodiment.

[0013] FIG. 2B is a schematic cross-sectional view illustrating positions of sheets in a case where a double-side printing job is performed in the image forming system of the first embodiment.

[0014] FIG. 3 is a schematic cross-sectional view illustrating a sheet feeding apparatus of the first embodiment.

[0015] FIG. 4 is a block diagram illustrating a control portion of the image forming system of the first embodiment.

[0016] FIG. 5 is a block diagram illustrating a control portion of the sheet feeding apparatus of the first embodiment.

[0017] FIG. 6 is a block diagram illustrating control functions of the sheet feeding apparatus of the first embodiment.

[0018] FIG. 7 is a table illustrating a conveyance-route management table of the sheet feeding apparatus of the first embodiment.

[0019] FIG. 8 is a diagram illustrating a sequence of conveyance control of the sheet feeding apparatus of the first embodiment.

[0020] FIG. 9 is a flowchart illustrating conveyance control of the sheet feeding apparatus of the first embodiment.

[0021] FIG. 10 is a flowchart illustrating multi-feeding detection control of the sheet feeding apparatus of the first embodiment.

[0022] FIG. 11 is a flowchart illustrating sheet-conveyance-path switch determination control of the sheet feeding apparatus of the first embodiment.

[0023] FIG. 12 is a flowchart illustrating switching-portion driving control of the sheet feeding apparatus of the first embodiment.

[0024] FIG. 13 is a time chart illustrating a relationship between a waveform detected by a multi-feeding detection sensor and the time.

[0025] FIG. 14A is a diagram illustrating a case where a switching portion can be switched after multi-fed sheets have passed through the switching portion in the sheet feeding apparatus.

[0026] FIG. 14B is a diagram illustrating a case where the switching portion cannot be switched after multi-fed sheets have passed through the switching portion in the sheet feeding apparatus.

[0027] FIG. 15 is a schematic cross-sectional view illustrating a sheet feeding apparatus of a second embodiment.

[0028] FIG. 16 is a flowchart illustrating conveyance control of the sheet feeding apparatus of the second embodiment.

DESCRIPTION OF THE EMBODIMENTS

First Embodiment

[0029] Hereinafter, a first embodiment for embodying the present invention will be described with reference to the accompanying drawings. In the first embodiment, the description will be made for a case where an ink-jet recording system is used as an image forming system 1.

Ink-Jet Recording System

[0030] First, a schematic configuration of the image forming system 1 of the first embodiment will be described with reference to FIGS. 1A and 1B. FIG. 1A is a block diagram illustrating a schematic configuration of the image forming system of the first embodiment. FIG. 1B is a schematic cross-sectional view illustrating the schematic configuration of the image forming system of the first embodiment. The image forming system 1 is a sheet-fed ink-jet recording system that forms an ink image on a sheet or a paper sheet (that is, produces a recorded product) by using two liquids: reaction liquid and ink. In the present embodiment, the sheet is a plain paper sheet, a thin paper sheet, a thick paper sheet, or a coated paper sheet. However, the sheet is not limited to the above-described sheets, and may be another sheet or a sheet made of a material other than paper. In addition, in the present embodiment, feeding of sheets is referred to as sheet feeding, and discharging of sheets is referred to as sheet discharging. The material of the sheet is not necessarily limited to paper.

[0031] As illustrated in FIG. 1A, the image forming system 1 is constituted by six units: a sheet feeding portion 100, a print portion 200 that serves as an image forming apparatus, a fixing portion 300, a cooling portion 400, a reversing portion 500, and a sheet discharging portion 600.

[0032] As illustrated in FIG. 1B, in the sheet feeding portion 100 that serves as a feeding portion that feeds sheets in the present embodiment, three sheet feeding apparatus 110, sheet feeding apparatus 120, and sheet feeding apparatus 130 that serve as sheet conveyance apparatuses are connected (linked) with each other. The sheet feeding apparatus 110 is a unit that includes sheet feeding trays, and includes a first sheet-feeding tray 111, a second sheet-feeding tray 112, and a third sheet-feeding tray 113 that serve as a three-tier support portion. The first sheet-feeding tray 111, the second sheet-feeding tray 112, and the third sheet-feeding tray 113 store various types of

printing sheets used in the print process. The sheet feeding apparatus **110** further includes a sheet discharging port **115** and a sheet discharging tray **119** that can be used as a destination to which an error sheet is sent. The error sheet is a sheet that cannot be normally conveyed to the print portion **200** when a jam occurs, a multi-fed sheet, or the like. Similarly, the sheet feeding apparatus **120** includes a first sheet-feeding tray **121**, a second sheet-feeding tray **122**, a third sheet-feeding tray **123**, a sheet discharging port **125**, and a sheet discharging tray **129**. The first sheet-feeding tray **121**, the second sheet-feeding tray **122**, and the third sheet-feeding tray **123** constitute a three-tier support portion. Furthermore, the sheet feeding apparatus **130** includes a first sheet-feeding tray **131**, a second sheet-feeding tray **132**, a third sheet-feeding tray **133**, a sheet discharging port **135**, and a sheet discharging tray **139**. The first sheet-feeding tray **131**, the second sheet-feeding tray **132**, and the third sheet-feeding tray **133** constitute a three-tier support portion.

[0033] The print portion **200** is a unit that performs printing on a sheet, which is used for printing a print image. The print portion **200** includes an ink-jet head **201** that serves as an image forming portion. The ink-jet head **201** constitutes an image forming portion that forms an image by performing a recording process (printing). In the recording process, the ink is ejected from a plurality of recording heads to a sheet that is being conveyed. The ink is ejected from above the sheet, and adheres to the sheet.

[0034] The fixing portion **300** is a unit that performs fixing control for fixing an image printed by the print portion **200**, to a sheet. The fixing portion **300** includes a drying module **310** and a fixing module **320**. The drying module **310** increases the fixability between the sheet and ink by blowing warm air on the sheet and reducing the liquid component of the ink adhered to the sheet. The fixing module **320** fixes the print image to the sheet by heating the sheet, by using a plurality of heater units. The fixing portion **300** includes a sheet discharging port **301** that can be used as a destination to which an error sheet is sent.

[0035] The cooling portion **400** is a unit that performs temperature control for returning the temperature of the sheet, heated by the fixing portion **300**, to a normal temperature by cooling the sheet. The cooling portion **400** cools the sheet by using a plurality of fan units.

[0036] The reversing portion **500** is a unit that performs reverse control of a sheet by switch-backing the sheet that is being conveyed. For stacking the sheet on the next-stage sheet discharging portion **600**, the reversing portion **500** performs the reverse control for switching the state of the sheet between a state where the sheet is to be stacked with a printed surface serving as an upper surface, and a state where the sheet is to be stacked with the printed surface serving as a lower surface. The reversing portion **500** includes a sheet discharging port **501** that can be used as a destination to which an error sheet is sent.

[0037] The sheet discharging portion **600** is a unit that stacks printed sheets and performs sheet discharging control. In the sheet discharging portion **600**, three sheet discharging apparatus **610**, sheet discharging apparatus **620**, and sheet discharging apparatus **630** are connected (linked) with each other. The sheet discharging apparatus **610** includes a sheet discharging stacker **611** that supports and stacks printed sheets. In addition, the sheet discharging apparatus **610** includes a sheet discharging port **612** that can be used as a destination to which a printed sheet or an error sheet is sent. Similarly, the sheet discharging apparatus **620** includes a sheet discharging stacker **621** and a sheet discharging port **622**, and the sheet discharging apparatus **630** includes a sheet discharging stacker **631** and a sheet discharging port **632**.

[0038] The image forming system **1** is connected with a print server **70**, and a print job is sent from the print server **70**. Furthermore, the print server **70** can be used for checking the state of the image forming system **1**, monitoring the print job, and performing the maintenance control. Thus, a user can operate the whole of various functions of the image forming system **1**.

[0039] The ink-jet head **201** of the print portion **200** is a precision portion that requires precision control. For example, if a plurality of sheets is multi-fed to the ink-jet head **201** in a state where the sheets overlap with each other, the distance between the ink-jet head **201** and the multi-fed sheets

cannot be kept properly, and the sheets may contact the ink-jet head **201**. In this case, the ink-jet head **201** may be damaged by the sheets. For preventing the damage of the ink-jet head **201**, it is necessary to remove the multi-fed sheets on the upstream side without conveying the multi-fed sheets to the print portion **200**.

Conveyance of Sheets in Image Forming System

[0040] Next, conveyance of sheets in the image forming system **1** will be described with reference to FIGS. **2A** and **2B**. FIG. **2A** is a schematic cross-sectional view illustrating positions of sheets in a case where a single-side printing job is performed in the image forming system of the first embodiment. FIG. **2B** is a schematic cross-sectional view illustrating positions of sheets in a case where a double-side printing job is performed in the image forming system of the first embodiment.

[0041] As illustrated in FIG. **2A**, in the execution of a print process of the single-side printing job, twenty sheets **St1** to **St20** are conveyed in the image forming system **1**. Specifically, if a sheet is fed from the sheet feeding tray **131** of the sheet feeding apparatus **130** of the sheet feeding portion **100**, the sheet is conveyed from the sheet feeding portion **100** to the print portion **200** for printing, and then conveyed sequentially through the units: the fixing portion **300**, the cooling portion **400**, and the reversing portion **500**. The sheet conveyed from the reversing portion **500** is stacked on the sheet discharging stacker **611** of the sheet discharging apparatus **610** of the sheet discharging portion **600**, as a printed sheet.

[0042] On the other hand, as illustrated in FIG. **2B**, in the execution of a print process of the double-side printing job, thirty-three sheets **St1** to **St33** are conveyed in the image forming system **1**. Specifically, if a sheet is fed from the sheet feeding tray **131** of the sheet feeding apparatus **130** of the sheet feeding portion **100**, the sheet is conveyed, as in the single-side printing job, from the sheet feeding portion **100** to the print portion **200** for printing, and then conveyed sequentially through the units: the fixing portion **300**, the cooling portion **400**, and the reversing portion **500**. A sheet with an image printed on the front side of the sheet is conveyed to a lower portion in the cooling portion **400**, then conveyed sequentially through a lower portion of the fixing portion **300** and a lower portion of the print portion **200**, and then returned to the sheet feeding apparatus **110** of the sheet feeding portion **100**. In the lower portion of the fixing portion **300** through which the sheet is conveyed, the front side and the back side of the sheet are reversed to each other. The sheet having been returned to the sheet feeding apparatus **110** of the sheet feeding portion **100** is conveyed to the print portion **200** for performing printing on the back side (i.e., a second side) of the sheet, and in the print portion **200**, the printing is performed on the back side of the sheet. After that, the sheet with images printed on both sides of the sheet is conveyed sequentially through the units: the fixing portion **300**, the cooling portion **400**, and the reversing portion **500**, and is stacked on the sheet discharging stacker **631** of the sheet discharging apparatus **630** of the sheet discharging portion **600**.

[0043] Note that in the present embodiment, the position of a sheet on which the control is performed is calculated, based on the time that has elapsed since the start of conveyance. That is, a time of a sheet calculated by a control portion of any one of units located upstream in the sheet conveyance direction is notified to a control portion of the unit, as a time at which the sheet will reach the entrance of the unit. In the present embodiment, the position (time) of a sheet is calculated, for example, by a control portion **150** of the sheet feeding apparatus **110** in the below-described control.

Configuration of Sheet Feeding Apparatus

[0044] Next, a configuration of the sheet feeding apparatus **110** of the sheet feeding portion **100** will be described with reference to FIG. **3**. FIG. **3** is a schematic cross-sectional view illustrating the sheet feeding apparatus of the first embodiment. Since the three sheet-feeding apparatus **110**, sheet-feeding apparatus **120**, and sheet-feeding apparatus **130** of the sheet feeding portion **100** of the present embodiment have the same configuration, the sheet feeding apparatus **110** of the three

sheet-feeding apparatuses will be described, and the description of the sheet feeding apparatus **120** and the sheet feeding apparatus **130** will be omitted.

[0045] As illustrated in FIG. **3**, the sheet feeding apparatus **110** includes the first to third sheet-feeding trays **111** to **113**, which serve as a three-tier support portion. As described above, the first to third sheet-feeding trays **111** to **113** stack and store various types of printing (white) sheets used in the print process. Above each of the first sheet-feeding tray **111**, the second sheet-feeding tray **112**, and the third sheet-feeding tray **113**, a corresponding one of sheet-feeding separation portion **111A**, sheet-feeding separation portion **112A**, and sheet-feeding separation portion **113A** is disposed. Each of the sheet-feeding separation portions serves as a feeding portion that feeds sheets supported by a corresponding sheet feeding tray, one by one. The sheet-feeding separation portion **111A**, the sheet-feeding separation portion **112A**, and the sheet-feeding separation portion **113A** respectively include sheet feeding belt **111a**, sheet feeding belt **112a**, and sheet feeding belt **113a**. Each of the sheet feeding belts is driven by a motor (not illustrated), and thereby feeds the uppermost sheet floated by a fan (not illustrated) while sticking to the sheet. In addition, the sheet-feeding separation portion **111A**, the sheet-feeding separation portion **112A**, and the sheet-feeding separation portion **113A** respectively include separation roller pair **111b**, separation roller pair **112b**, and separation roller pair **113b**. Each of the separation roller pairs separates one sheet from the other if sheets are multi-fed by a corresponding one of the sheet feeding belt **111a**, the sheet feeding belt **112a**, and the sheet feeding belt **113a**. Each of the separation roller pair **111b**, the separation roller pair **112b**, and the separation roller pair **113b** is constituted by a common feed roller and a common retard roller, but may have any configuration. For example, another component, such as a separation pad, may be used in each of the separation roller pairs.

[0046] For detecting the multi-feeding of sheets, multi-feeding detection sensor **SN11**, multi-feeding detection sensor **SN12**, and multi-feeding detection sensor **SN13** are disposed respectively downstream of the separation roller pair **111b**, the separation roller pair **112b**, and the separation roller pair **113b**, in a sheet conveyance direction V. Each of the multi-feeding detection sensor **SN11**, the multi-feeding detection sensor **SN12**, and the multi-feeding detection sensor **SN13** starts sensing at the leading edge of a fed sheet, and detects whether a plurality of sheets is fed simultaneously, and how much one sheet is shifted from another if the multi-feeding occurs. The detection of the multi-feeding performed by the multi-feeding detection sensor **SN11**, the multi-feeding detection sensor **SN12**, and the multi-feeding detection sensor **SN13** will be described in detail below.

[0047] Next, the conveyance path of sheets in the sheet feeding apparatus **110** will be described. The sheet feeding apparatus **110** includes a horizontal-path entrance **EN2** that receives a sheet conveyed from an apparatus (i.e., the sheet feeding apparatus **120**) disposed upstream in the sheet conveyance direction. In addition, the sheet feeding apparatus **110** includes a sheet discharging port **EX1** that discharges a sheet to an apparatus (i.e., the print portion **200**) disposed downstream in the sheet conveyance direction V. Furthermore, the sheet feeding apparatus **110** includes a double-side-printing-path entrance **EN1** that receives a sheet conveyed from the above-described lower portion of the print portion **200** to the sheet feeding apparatus **110** for the double-side printing. In addition, the sheet feeding apparatus **110** includes a sheet discharging tray **119** that serves as a discharging support portion. As described in detail below, the sheet discharging tray **119** is a destination to which an error sheet is sent and discharged. That is, a sheet that becomes unacceptable or invalid due to the multi-feeding or the like is discharged to the sheet discharging tray **119**, and stacked on and supported by the sheet discharging tray **119**. Thus, the sheet feeding apparatus **110** includes the sheet discharging port **115** that can discharge the sheet to the sheet discharging tray **119**. Similarly, the sheet feeding apparatus **120** and the sheet feeding apparatus **130** respectively include the above-described sheet discharging port **125** and the sheet discharging port **135** (see FIG. **1B**), from which sheets can be discharged to the sheet discharging tray **129** and the sheet discharging tray **139**.

[0048] The sheet feeding apparatus **110** includes a plurality of conveyance paths that causes the

horizontal-path entrance EN2, the sheet discharging port EX1, the double-side-printing-path entrance EN1, the sheet discharging port 115, and the first to third sheet-feeding trays 111 to 113 to communicate with each other, and that guides a sheet. In addition, in the conveyance paths, a plurality of conveyance roller pairs is disposed for conveying sheets. The conveyance roller pairs constitute a conveyance portion 110A that conveys sheets.

[0049] Specifically, the sheet feeding apparatus 110 includes a first horizontal-conveyance path Ps11, a second horizontal-conveyance path Ps12 that serves as a first conveyance path, and a third horizontal-conveyance path Ps13 that serves as a second conveyance path. The first horizontal-conveyance path Ps11, the second horizontal-conveyance path Ps12, and the third horizontal-conveyance path Ps13 are horizontal paths that cause the horizontal-path entrance EN2 and the sheet discharging port EX1 to communicate with each other. In addition, the first horizontal-conveyance path Ps11, the second horizontal-conveyance path Ps12, and the third horizontal-conveyance path Ps13 are disposed, one adjacent to another on a straight line in a horizontal direction. In addition, the sheet feeding apparatus 110 includes an error sheet-discharging path Ps14 that serves as a third conveyance path. The error sheet-discharging path Ps14 is branched from an error-bin branch point P5 positioned between the second horizontal-conveyance path Ps12 and the third horizontal-conveyance path Ps13, and guides a sheet to the sheet discharging port 115. At the error-bin branch point P5, a switching portion FL1 is disposed. The switching portion FL1 can be switched between a first position and a second position by pivoting the switching portion FL1. In the first position, the switching portion FL1 guides a sheet conveyed in the second horizontal-conveyance path Ps12 by the conveyance portion 110A, to the third horizontal-conveyance path Ps13. In the second position, the switching portion FL1 guides a sheet conveyed in the second horizontal-conveyance path Ps12 by the conveyance portion 110A, to the error sheet-discharging path Ps14.

[0050] In addition, the sheet feeding apparatus 110 includes a duplex conveyance path Ps15. The duplex conveyance path Ps15 causes the double-side-printing-path entrance EN1 and a horizontal-path meeting point P4 positioned between the first horizontal-conveyance path Ps11 and the second horizontal-conveyance path Ps12, to communicate with each other, and guides a sheet. In addition, the sheet feeding apparatus 110 includes a tray conveyance path Ps18 that causes the third sheet-feeding tray 113 and a third-sheet-feeding-tray meeting point P1 to communicate with each other, and guides a sheet. In addition, the sheet feeding apparatus 110 includes a tray conveyance path Ps17 that causes the second sheet-feeding tray 112 and a second-sheet-feeding-tray meeting point P2 to communicate with each other, and guides a sheet. In addition, the sheet feeding apparatus 110 includes a tray conveyance path Ps16 that causes the first sheet-feeding tray 111 and a first-sheet-feeding-tray meeting point P3 to communicate with each other, and guides a sheet.

[0051] In the sheet feeding apparatus 110 configured as described above, a sheet fed from the first sheet-feeding tray 111 by the sheet-feeding separation portion 111A is conveyed to the sheet discharging port EX1 through the tray conveyance path Ps16, the duplex conveyance path Ps15, the second horizontal-conveyance path Ps12, and the third horizontal-conveyance path Ps13. Similarly, a sheet fed from the second sheet-feeding tray 112 by the sheet-feeding separation portion 112A is conveyed to the sheet discharging port EX1 through the tray conveyance path Ps17, the duplex conveyance path Ps15, the second horizontal-conveyance path Ps12, and the third horizontal-conveyance path Ps13. Similarly, a sheet fed from the third sheet-feeding tray 113 by the sheet-feeding separation portion 113A is conveyed to the sheet discharging port EX1 through the tray conveyance path Ps18, the duplex conveyance path Ps15, the second horizontal-conveyance path Ps12, and the third horizontal-conveyance path Ps13.

[0052] In addition, a sheet conveyed from the above-described sheet feeding apparatus 120 or the sheet feeding apparatus 130 (see FIG. 1B) is conveyed to the sheet discharging port EX1 through the horizontal-path entrance EN2, the first horizontal-conveyance path Ps11, the second horizontal-conveyance path Ps12, and the third horizontal-conveyance path Ps13. In addition, a sheet

conveyed from the above-described print portion **200** (see FIG. **1B**) is conveyed to the sheet discharging port **EX1** through the double-side-printing-path entrance **EN1**, the duplex conveyance path **Ps15**, the second horizontal-conveyance path **Ps12**, and the third horizontal-conveyance path **Ps13**.

[0053] The above-described switching portion **FL1** is driven by a switching-portion driving motor (not illustrated). When power is turned on or a print job is started, the switching portion **FL1** is initialized so as to guide a sheet located in the second horizontal-conveyance path **Ps12**, to the third horizontal-conveyance path **Ps13** (i.e., the sheet discharging port **EX1**). If the multi-feeding detection sensor **SN11**, the multi-feeding detection sensor **SN12**, or the multi-feeding detection sensor **SN13**, that serves as a multi-feeding detection portion detects the multi-feeding of sheets in the tray conveyance path **Ps16**, the tray conveyance path **Ps17**, or the tray conveyance path **Ps18**, the switching portion **FL1** is switched from the first position to the second position, and the switching portion **FL1** guides a sheet located in the second horizontal-conveyance path **Ps12**, to the error sheet-discharging path **Ps14** (i.e., the sheet discharging port **115**). The detailed description of the switch control of the switching portion **FL1** will be made below.

Control Portion of Image Forming System

[0054] Next, a control portion **10** of the image forming system **1** will be described with reference to FIG. **4**. FIG. **4** is a block diagram illustrating the control portion of the image forming system of the first embodiment.

[0055] As illustrated in FIG. **4**, the image forming system **1** includes the control portion **10** that is a controller unit. The control portion **10** controls sending and receiving sensor signals and device information, and controls executing various processes of control. For example, a program in which processes are written is stored in a ROM **13** or an HDD **14**, and installed in the image forming system **1**. A CPU **11** of the control portion **10** reads the program stored in the ROM **13** or the HDD **14**, from the ROM **13** or the HDD **14**; stores the program in a RAM **12**; and executes the program. In addition, the CPU **11** collectively controls the devices connected to a system bus **15**. The RAM **12** functions as a main memory or a work memory of the CPU **11**. The ROM **13** stores a boot program executed when power is turned on, and the HDD **14** stores an operating system and a control program of the system. The HDD **14** is also used for storing large data temporarily or for a long time.

[0056] A Network **16** is connected to a local area network **17**, and sends/receives data or device information to/from an external apparatus. The program may be installed in the ROM **13** or the HDD **14** via the Network **16**. A device I/F **18** is an interface portion between the CPU **11** and a printer engine **19**, and sends/receives signals for operating or referring to various motors, sensors, and the ink-jet head connected to the printer engine **19**. The printer engine **19** is an ink-jet output device that conveys a printing sheet, depending on the control from the control portion **10**, and that outputs an image sent from the print server **70**, onto a printing sheet by controlling the ink-jet head **201** (see FIG. **1B**). A head control portion **20** controls the ink-jet head **201** of the printer engine **19** via the device I/F **18**, and performs the control, such as ink-ejection control for the image formation and head retraction control performed when an improper sheet is detected. A conveyance control portion **21** controls conveyance motors and sensors for feeding a printing sheet from the sheet feeding portion **100**, conveying the sheet in synchronization with an ink ejection timing of the ink-jet head **201**, and discharging the sheet to the sheet discharging portion **600**.

[0057] Note that the description has been made for the case where the control portion **10** (i.e., a controller unit) is a single unit. However, a plurality of controller units may be used, separated from each other, and each of the sheet feeding portion **100**, the print portion **200**, the fixing portion **300**, the cooling portion **400**, the reversing portion **500**, and the sheet discharging portion **600** may include a corresponding one of the controller units. In this case, a specific controller unit collectively controls the other controller units.

Control Portion of Sheet Feeding Apparatus

[0058] Next, a configuration of a control portion **150** of the sheet feeding apparatus **110** will be described with reference to FIG. 5. FIG. 5 is a block diagram illustrating the control portion of the sheet feeding apparatus of the first embodiment. The description will be made for a case where the sheet feeding apparatus **110**, the sheet feeding apparatus **120**, and the sheet feeding apparatus **130** of the sheet feeding portion **100** include corresponding control portions (i.e., controller units) separated from each other, and where the control portion **150** of the sheet feeding apparatus **110** is one of the control portions. That is, each of the sheet feeding apparatus **120** and the sheet feeding apparatus **130** includes a control portion identical to the control portion **150**. In this configuration, even in a case where the sheet feeding portion **100** includes three apparatuses (units) identical to each other, connected in series with each other, and serving as sheet feeding units (that is, three sheet feeding apparatuses identical to each other and connected in series with each other), the system can be built easily.

[0059] As illustrated in FIG. 5, the sheet feeding apparatus **110** includes the control portion **150** that is a controller unit. The control portion **150** controls sending and receiving sensor signals and device information, and controls executing various processes of control. For example, a program in which processes are written is stored in a ROM **153** or an HDD **154**, and installed in the sheet feeding apparatus **110**. A CPU **151** reads the program stored in the ROM **153** or the HDD **154**, from the ROM **153** or the HDD **154**; stores the program in a RAM **152**; and executes the program. In addition, the CPU **151** collectively controls the devices connected to a system bus **155**. The RAM **152** functions as a main memory or a work memory of the CPU **151**. The ROM **153** stores a boot program executed when power is turned on, and the HDD **154** stores an operating system and a control program of the apparatus. The HDD **154** is also used for storing large data temporarily or for a long time.

[0060] A Network **156** is connected to a local area network **157**, and sends/receives data or device information to/from an external apparatus. The program may be installed in the ROM **153** or the HDD **154** via the Network **156**. A unit I/F **158** is an interface portion between the CPU **151** and an apparatus disposed adjacent to the sheet feeding apparatus **110**. In the present embodiment, the unit I/F **158** is connected to the sheet feeding apparatus **120** (i.e., the control portion of the sheet feeding apparatus **120**) connected to the sheet feeding apparatus **110**. A controller I/F **160** is an interface portion between the controller and the print portion **200**, and is connected with the print portion **200** of the image forming system **1** (i.e., the control portion of the print portion **200**). A device control portion **161** controls sheet feeding fans for feeding printing sheets from the first to third sheet-feeding trays **111** to **113**, and controls conveyance motors and sensors for conveying sheets to the downstream apparatus or the sheet discharging tray **119** (see FIG. 3). As described above, each of the multi-feeding detection sensor SN**11**, the multi-feeding detection sensor SN**12**, and the multi-feeding detection sensor SN**13** detects the multi-feeding of sheets fed from a corresponding one of the sheet-feeding separation portion **111A**, the sheet-feeding separation portion **112A**, and the sheet-feeding separation portion **113A**. A switching-portion driving motor **162** (FLAPPER DRIVING MOTOR) drives the switching portion FL**1** for switching the above-described position of the switching portion FL**1**.

Control Functions of Sheet Feeding Apparatus

[0061] Next, control functions achieved by the control portion **150** of the above-described sheet feeding apparatus **110** performing various types of control (processes) will be described with reference to FIG. 6. FIG. 6 is a block diagram illustrating the control functions of the sheet feeding apparatus of the first embodiment. Note that the control portion of each of the sheet feeding apparatus **120** and the sheet feeding apparatus **130** also has the same control functions as those illustrated in FIG. 6. Each component illustrated in FIG. 6 and serving as a software module functions such that the component is stored in the HDD **154**, as a program, then loaded into the RAM **152**, and then executed by the CPU **151**.

[0062] A sheet-feeding-apparatus management portion SS**1** sends/receives print-job information

and instructions or responses to/from the print portion **200** via the controller I/F **160**, and collectively controls the components of the sheet feeding apparatus **110**. Upon receiving a request for conveying a sheet (i.e., a request for feeding a sheet or a request for conveying a sheet in the double-side printing), the sheet-feeding-apparatus management portion **SS1** sends an instruction for conveying a sheet, to a conveyance control portion **SS2**. The conveyance control portion **SS2** sends an instruction to a sheet-feeding control portion **SS3**, a multi-feeding detection portion **SS4**, or a conveyance-path switch control portion **SS5** in accordance with a conveyance route of the above-described conveyance paths. That is, while sending an instruction to each of the control portions, the conveyance control portion **SS2** performs the control for conveying a sheet to be fed or a sheet conveyed from one adjacent apparatus, to another adjacent apparatus (i.e., the print portion **200**) disposed downstream in the conveyance direction **V**, or to the sheet discharging port **115** in accordance with a sheet conveyance time.

[0063] The sheet-feeding control portion **SS3** controls the sheet-feeding separation portion **111A**, the sheet-feeding separation portion **112A**, and the sheet-feeding separation portion **113A**; and feeds sheets, one by one, stacked on each of the sheet feeding tray **111**, the sheet feeding tray **112**, and the sheet feeding tray **113**. The multi-feeding detection portion **SS4** causes the multi-feeding detection sensor **SN11**, the multi-feeding detection sensor **SN12**, and the multi-feeding detection sensor **SN13** to sense sheets fed from the corresponding sheet feeding tray **111**, the corresponding sheet feeding tray **112**, and the corresponding sheet feeding tray **113**. That is, each of the multi-feeding detection sensor **SN11**, the multi-feeding detection sensor **SN12**, and the multi-feeding detection sensor **SN13** detects whether sheets are multi-fed, and how much a plurality of sheets that overlap with each other is shifted from each other in the conveyance in the conveyance direction **V** if the sheets are multi-fed. The conveyance-path switch control portion **SS5** switches the destination of sheets to the sheet discharging port **115** by switching the switching portion **FL1** in a case where the multi-feeding of sheets has been detected or the sheets that are being conveyed have become unacceptable due to another error. In this case, the sheets are determined as error sheets. The switching portion **FL1** is switched by driving the switching-portion driving motor **162**.

Conveyance-Route Management Table

[0064] Next, a conveyance-route management table included in the control portion **150** of the sheet feeding apparatus **110** will be described with reference to FIG. 7. FIG. 7 is a table illustrating the conveyance-route management table of the sheet feeding apparatus of the first embodiment.

[0065] In the conveyance of a sheet, there is a case where the sheet is conveyed from the conveyance start position of each component to the sheet discharging port (an error bin) past a branch position and a meeting position. For managing the time at which the sheet passes the branch position and the meeting position, in the conveyance-route management table, each position and a corresponding passage time are recorded, associated with each other, as illustrated in FIG. 7. Note that in the present embodiment, the position of a sheet which is conveyed and on which the control is performed is calculated, based on the time that has elapsed since the start of conveyance.

[0066] The conveyance-route management table contains information on the Index-No, information on the branch or meeting position, and information on the conveyance time from a conveyance start position to a branch or meeting position. The Index-No is an identifier for uniquely identifying the conveyance route information. The branch or meeting position is the name of the branch position (**P5**) and the meeting positions (**P1**, **P2**, **P3**, and **P4**) (see FIG. 3). The conveyance time from a conveyance start position to a branch or meeting position is a conveyance time taken from a conveyance start position (0 mm) that is the first sheet-feeding tray **111**, the second sheet-feeding tray **112**, the third sheet-feeding tray **113**, the double-side-printing-path entrance **EN1**, or the horizontal-path entrance **EN2**. That is, the conveyance time from a conveyance start position to a branch or meeting position is a time taken for a sheet to move from the conveyance start position to the branch or meeting position, which are positioned on the conveyance path for discharging the sheet to the target sheet-discharging destination. That is, the

conveyance-route management table contains information on the time at which a sheet conveyed from each conveyance start position reaches a branch or meeting position. Thus, the time at which a sheet passes a point on the conveyance path can be calculated from the information contained in the conveyance-route management table, and from the time which is notified from the control portion **10** of the print portion **200** and at which the sheet is located in the conveyance start position.

Sequence for Feeding and Conveying Sheets

[0067] Next, an overall sequence of the sheet feeding apparatus **110** used for feeding sheets, one by one, stacked on the first to third sheet-feeding trays **111** to **113** will be described with reference to FIG. **8**. FIG. **8** is a diagram illustrating the sequence of conveyance control of the sheet feeding apparatus of the first embodiment.

[0068] As illustrated in FIG. **8**, if the sheet-feeding-apparatus management portion **SS1** receives a conveyance request from the print portion **200**, the sequence is started after the sheet-feeding-apparatus management portion **SS1** initializes the switching portion **FL1** (that is, moves the switching portion **FL1** to a home position at which the switching portion **FL1** guides a sheet to the sheet discharging port **EX1**) and drives the fans for feeding a sheet. The conveyance request notified contains information on a sheet feeding tray from which a sheet is to be fed, and information on the sheet-conveyance start time.

[0069] In Step **S101**, the sheet-feeding-apparatus management portion **SS1** notifies the conveyance control portion **SS2** of an instruction for feeding and conveying a sheet. In Step **S102**, the conveyance control portion **SS2** notifies the sheet-feeding control portion **SS3** of the instruction for feeding and conveying a sheet. When the conveyance control portion **SS2** notifies the sheet-feeding control portion **SS3** of the instruction for feeding and conveying a sheet, the conveyance control portion **SS2** also notifies the sheet-feeding control portion **SS3** of the sheet-conveyance start time.

[0070] In Step **S103**, the sheet-feeding control portion **SS3** causes the sheet-feeding separation portion of a specified sheet feeding tray to perform the sheet feeding control. That is, the sheet-feeding separation portion rotates the fan for floating a sheet, and performs the sheet feeding operation by driving the sheet feeding belt if the sheet-conveyance start time is reached. In Step **S104**, the sheet-feeding control portion **SS3** notifies the conveyance control portion **SS2** of the completion of feeding a sheet. The notification may be performed after Step **S105**.

[0071] In Step **S105**, the conveyance control portion **SS2** notifies the multi-feeding detection portion **SS4** of a multi-feeding-detection start instruction. When the conveyance control portion **SS2** notifies the multi-feeding detection portion **SS4** of the multi-feeding-detection start instruction, the conveyance control portion **SS2** also notifies the multi-feeding detection portion **SS4** of the time at which the leading edge of a fed sheet will pass the multi-feeding detection sensor. In Step **S106**, the multi-feeding detection portion **SS4** performs the below-described multi-feeding detection control.

[0072] In Step **S107**, the multi-feeding detection portion **SS4** notifies the conveyance control portion **SS2** of a result of the multi-feeding detection. The notification contains information on whether the multi-feeding has occurred, and an elapsed time in which the leading edge of the sheet has moved when the multi-feeding is detected. In Step **S108**, the conveyance control portion **SS2** performs below-described sheet-conveyance-path switch determination.

[0073] In Step **S109**, the conveyance control portion **SS2** notifies the conveyance-path switch control portion **SS5** of a sheet-conveyance-path switch instruction, if necessary. When the conveyance control portion **SS2** notifies the conveyance-path switch control portion **SS5** of the sheet-conveyance-path switch instruction, the conveyance control portion **SS2** notifies the conveyance-path switch control portion **SS5** of a switch start time of the switching portion **FL1** and a return start time that serves as a first time. In Step **S110**, the conveyance-path switch control portion **SS5** performs below-described switching-portion driving control (FLAPPER DRIVING CONTROL). In Step **S111**, the conveyance-path switch control portion **SS5** notifies the

conveyance control portion SS2 of the completion of switching the sheet conveyance path.

[0074] If the conveyance of the sheet to the sheet discharging port EX1 or the discharging of the sheet to the sheet discharging port 115 is completed, the conveyance control portion SS2 notifies the sheet-feeding-apparatus management portion SS1 of the completion of feeding and conveying the sheet, in Step S112. In the above-described sequence, the processes for feeding and conveying sheets are performed.

Control of Sheet Feeding Apparatus of First Embodiment

[0075] Hereinafter, the control of the sheet feeding apparatus of the present embodiment will be described in detail with reference to a flowchart. The following description will be made for a case where the control is performed by the CPU 151 of the control portion 150 of the sheet feeding apparatus 110. However, the control may be performed by a CPU of any one of the control portions of the sheet feeding apparatuses. In particular, the control may be performed by the CPU 11 of the control portion 10 of the image forming system 1. In addition, in the following description, the first sheet that a sheet feeding tray starts to feed is referred to as a sheet S1, the second sheet following the first sheet and multi-fed is referred to as a sheet S2, and a sheet conveyed from an upstream apparatus so as to follow the sheet S1 or S2 is referred to as a sheet S3.

Conveyance Control

[0076] First, the conveyance control of the sheet feeding apparatus of the first embodiment will be described with reference to FIG. 9. FIG. 9 is a flowchart illustrating the conveyance control of the sheet feeding apparatus of the first embodiment. In the following description of the flowchart, an instruction or a notification is equivalent to setting ON or OFF in a flag (involved with a corresponding process) of the control portion 150 of the sheet feeding apparatus 110 or a control portion of another apparatus. If the flag is set in accordance with the instruction or the notification, any process involved with the flag that has been set can be started, or any image can be displayed by a display (not illustrated).

[0077] As described above, if the conveyance control portion SS2 receives, in Step S101, an instruction for feeding and conveying a sheet, from the sheet-feeding-apparatus management portion SS1, the conveyance control of the sheet feeding apparatus is started. As illustrated in FIG. 9, the CPU 151 then drives a conveyance roller of each component of the conveyance portion 110A (S11). Then the CPU 151 determines whether the instruction for feeding and conveying a sheet is an instruction for feeding a sheet from any one of the first sheet-feeding tray 111, the second sheet-feeding tray 112, and the first sheet-feeding tray 113, or an instruction for receiving a sheet conveyed from an upstream apparatus (i.e., another sheet feeding apparatus or the print portion 200) disposed upstream in the conveyance direction V (S12). The CPU 151 proceeds to Step S13 if the instruction for feeding and conveying a sheet is an instruction for feeding a sheet from any one of the first sheet-feeding tray 111, the second sheet-feeding tray 112, and the first sheet-feeding tray 113, or proceeds to Step S23 if the instruction for feeding and conveying a sheet is an instruction for receiving a sheet conveyed from the upstream apparatus.

[0078] If the CPU 151 proceeds to Step S13, then the CPU 151 notifies the sheet-feeding control portion SS3 of a sheet-feeding start instruction. When the CPU 151 notifies the sheet-feeding control portion SS3 of the sheet-feeding start instruction, the CPU 151 notifies the sheet-feeding control portion SS3 (i.e., the sheet-feeding separation portion) of information that, for example, contains the data: “cassette ID: second sheet-feeding cassette, sheet-conveyance start time: 12:10:30 and 000 milliseconds” (the sheet-conveyance start time, 12:10:30 and 000 milliseconds, is hereinafter abbreviated as 30.000). Upon receiving the sheet-feeding start instruction, the sheet-feeding control portion SS3 performs a process for feeding a sheet, and starts to feed the sheet from the second sheet-feeding tray 112 at the sheet-conveyance start time of 30.000.

[0079] Then the CPU 151 notifies the multi-feeding detection portion SS4 (i.e., the multi-feeding detection sensor) of a multi-feeding-detection start instruction (S14). When the CPU 151 notifies the multi-feeding detection portion SS4 of the multi-feeding-detection start instruction, the CPU

151 notifies the multi-feeding detection portion **SS4** of information that, for example, contains the data: “sensor ID: multi-feeding detection sensor **SN12**, detection start time: 30.150, detection end time: 30.570” if an A3-size sheet (whose length is 420 mm) is conveyed at a speed of 1000 mm/s. The detection start time is determined as described above because the leading edge of the fed sheet passes the multi-feeding detection sensor **SN12** when 150 msec has elapsed since the sheet conveyance time. The detection end time is determined, based on a time of 420 msec in which the sheet having a length of 420 mm passes through the multi-feeding detection sensor **SN12** (the time of period in which the detection is performed may be determined based on the sheet length, or may be determined based on another parameter). Upon receiving the multi-feeding-detection start instruction, the multi-feeding detection portion **SS4** performs a process for detecting the multi-feeding, and starts the detection of the multi-feeding at the detection start time of 30.150. The detection of the multi-feeding will be described in detail below.

[0080] Then the CPU **151** determines whether a sheet-conveyance-path switch determination time is reached, for conveying the sheet **S1** fed from the sheet feeding tray (**S15**). In the present embodiment, the CPU **151** determines that the sheet-conveyance-path switch determination time is reached, when the leading edge of the sheet **S1** reaches the horizontal-path meeting point (**P4**) (that is, at a time at which 605 msec has elapsed since the start of conveyance of the sheet **S1** from the second sheet-feeding cassette) (see FIG. 7). Thus, the CPU **151** determines whether the arrival time of 30.605 is reached. If the arrival time is not reached, then the CPU **151** repeats the process until the arrival time is reached (**S15**: No). If the arrival time is reached (**S15**: Yes), then the CPU **151** proceeds to Step **S16**. In Step **S16**, the CPU **151** performs sheet-conveyance-path switch determination. The sheet-conveyance-path switch determination will be described in detail below.

[0081] If the CPU **151** determines, based on the result of the sheet-conveyance-path switch determination, that it is necessary to switch the sheet conveyance path (**S17**: Yes), then the CPU **151** proceeds to Step **S18**. If the CPU **151** determines that it is unnecessary to switch the sheet conveyance path (**S17**: No), then the CPU **151** proceeds to Step **S27**. If the CPU **151** proceeds to Step **S18**, then the CPU **151** notifies the conveyance-path switch control portion **SS5** of a sheet-conveyance-path switch instruction. When the CPU **151** notifies the conveyance-path switch control portion **SS5** of the sheet-conveyance-path switch instruction, the CPU **151** notifies the conveyance-path switch control portion **SS5** of information that, for example, contains the data: “switch start time: 30.615, return start time: 31:145”. The data is created, depending on the result of the sheet-conveyance-path switch determination. Upon receiving the information, the conveyance-path switch control portion **SS5** switches the switching portion **FL1**.

[0082] Then the CPU **151** notifies the sheet-feeding-apparatus management portion **SS1** of the occurrence of multi-feeding (**S19**). In addition, the occurrence of multi-feeding is notified to the print portion **200** via the sheet-feeding-apparatus management portion **SS1**. The notification may be performed when the sheet is discharged to the sheet discharging port **115**. However, if the notification is performed earlier, the following sheet can be stopped earlier, and the operation of the print portion **200** can be stopped or changed earlier.

[0083] Then the CPU **151** determines whether the sheets (i.e., the multi-fed sheet **S1** and the multi-fed sheet **S2** in the present embodiment) are discharged to the sheet discharging port **115** (**S20**). The completion of discharging sheets may be determined based on time. In another case, a sensor for monitoring the conveyance of sheets may be disposed in the vicinity of the sheet discharging port **115**, and the sensor may monitor the passage of the trailing edge of the sheet **S2**. The sheet-discharging completion time can be calculated by adding the time at which the leading edge of the sheet **S1** reaches the sheet discharging port **115** (when 905 msec has elapsed since the start of conveyance of the sheet **S1** from the second sheet-feeding cassette) (see FIG. 7), with a time that corresponds to a length of a multi-fed sheet (including the below-described amount of shift). If the sheet discharging is not completed (**S20**: No), then the CPU **151** repeats the process until the sheet discharging is completed. If the sheet discharging is completed (**S20**: Yes), then the CPU **151**

proceeds to Step S21.

[0084] If the CPU 151 proceeds to Step S21, then the CPU 151 notifies the conveyance control portion SS2 of the completion of discharging the sheets to the error bin (i.e., the sheet discharging port 115). With this operation, the CPU 151 stops the sheet discharging roller (S22), and ends the conveyance control of the sheet feeding apparatus 110.

[0085] On the other hand, if the instruction for feeding and conveying a sheet is, in Step S12, an instruction for receiving a sheet conveyed from the above-described upstream apparatus, then the CPU 151 proceeds to Step S23. If the CPU 151 proceeds to Step S23, then the CPU 151 determines whether a sheet reception time is reached. The CPU 151 repeats the process if the sheet reception time is not reached (S23: No), or proceeds to Step S24 if the sheet reception time is reached (S23: Yes). If the CPU 151 proceeds to Step S24, then the CPU 151 receives the sheet S3 from the upstream apparatus.

[0086] Then the CPU 151 determines whether a passage time at which the sheet S3 received from the upstream apparatus passes the error-bin branch point (P5) is reached (S25). The CPU 151 repeats the process if the passage time is not reached (S25: No), or proceeds to Step S26 if the passage time is reached (S25: Yes).

[0087] In Step S26, the CPU 151 determines whether the direction of the conveyance path extends toward a downstream apparatus disposed downstream in the conveyance direction V (i.e., the print portion 200, the sheet feeding apparatus 110 if the sheet is fed from the sheet feeding apparatus 120, or the sheet feeding apparatus 120 if the sheet is fed from the sheet feeding apparatus 130). That is, the CPU 151 determines whether the switching portion FL1 is positioned in the first position that is an initial position. If the direction of the conveyance path extends toward the downstream apparatus (S26: Yes), then the CPU 151 proceeds to Step S27.

[0088] If it is not necessary to switch the sheet conveyance path in Step S17 (S17: No), or the direction of the conveyance path extends toward the downstream apparatus (S26: Yes), then the CPU 151 proceeds to Step S27. Then the CPU 151 determines whether the sheet (i.e., the sheet S1 or S3 in the present embodiment) is discharged to the downstream apparatus. The completion of discharging the sheet to the downstream apparatus may be determined based on time. In another case, a sensor for monitoring the conveyance of sheets may be disposed in the vicinity of the sheet discharging port EX1, and the sensor may monitor the passage of the trailing edge of the sheet S1 or S3. The sheet-discharging completion time can be calculated by adding the time at which the leading edge of the sheet reaches the sheet discharging port EX1 (when 705 msec has elapsed since the start of conveyance of the sheet S1 from the second sheet-feeding cassette, for example) (see FIG. 7), with a time that corresponds to a length of the sheet. If the sheet discharging is not completed (S27: No), then the CPU 151 repeats the process until the sheet discharging is completed. If the sheet discharging is completed (S27: Yes), then the CPU 151 proceeds to Step S28. If the CPU 151 proceeds to Step S28, then the CPU 151 notifies the conveyance control portion SS2 of the completion of normally delivering the sheet S1 or S3 to the downstream apparatus, and proceeds to Step S22. With this operation, the CPU 151 stops the sheet discharging roller (S22), and ends the conveyance control of the sheet feeding apparatus 110.

[0089] On the other hand, if the direction of the conveyance path does not extend toward the downstream apparatus in the above-described Step S26 (S26: No), that is, if the switching portion FL1 has been switched from the first position to the second position and the direction of the conveyance path extends toward the sheet discharging port 115, then the CPU 151 proceeds to Step S20. In this case, the preceding sheet S1 and S2 have been multi-fed, and the sheet S3 conveyed from the upstream apparatus has reached the switching portion FL1 before the switching portion FL1 is returned to the first position, because the amount of shift is large. Thus, in this case, the switching portion FL1 cannot be returned to the first position.

[0090] Also in a case where the CPU 151 proceeds to Step S20 in this manner, the CPU 151 determines whether the sheets (i.e., the sheets S1 to S3) are discharged to the sheet discharging port

115, as described above (**S20**). If the sheet discharging is not completed (**S20**: No), then the CPU **151** repeats the process until the sheet discharging is completed. If the sheet discharging is completed (**S20**: Yes), then the CPU **151** notifies the conveyance control portion **SS2** of the completion of discharging the sheets to the error bin (i.e., the sheet discharging port **115**). With this operation, the CPU **151** stops the sheet discharging roller (**S22**), and ends the conveyance control of the sheet feeding apparatus **110**.

[0091] Note that in the above-described conveyance control of the sheet feeding apparatus **110**, the plurality of conveyance rollers that serve as the conveyance portion **110A** is collectively driven or stopped. However, only necessary conveyance rollers may be driven or stopped in a sequential manner by calculating the time at which the sheet reaches a corresponding component.

Multi-Feeding Detection Control

[0092] Next, the multi-feeding detection control of the sheet feeding apparatus of the first embodiment will be described with reference to FIGS. **10** and **13**. FIG. **10** is a flowchart illustrating the multi-feeding detection control of the sheet feeding apparatus of the first embodiment. FIG. **13** is a time chart illustrating a relationship between a waveform detected by a multi-feeding detection sensor and the time.

[0093] First, the relationship between a waveform detected by the multi-feeding detection sensor and the time will be described. As illustrated in FIG. **13**, the second sheet **S2** following the preceding first sheet **S1** is conveyed, overlapped with the first sheet **S1**. That is, the second sheet **S2** is conveyed, multi-fed. In a case where the sheet **S1** and the sheet **S2** are conveyed, shifted from each other, a detection start time T_s is a time at which the leading edge of the sheet **S1** passes the multi-feeding detection sensor, and a multi-feeding occurrence time T_d is a time at which the leading edge of the sheet **S2** reaches the multi-feeding detection sensor. In a case where an ultrasonic multi-feeding-detection sensor is used, if the sheets are conveyed, shifted from each other, the waveform of an overlapping portion of the sheets detected by the sensor decreases in level. Thus, if the level of the waveform becomes lower than a predetermined detection threshold, the CPU **151** determines the occurrence of multi-feeding. A multi-feeding shift time t that corresponds to the amount of shift between the sheet **S1** and the sheet **S2** can be calculated by subtracting the detection start time T_s from the multi-feeding occurrence time T_d . In other words, the multi-feeding shift time t is a length of time from the time when the leading edge of the sheet **S1** is detected, to the time when the multi-feeding is detected. The multi-feeding shift time t corresponds to a conveyance time t' by which the conveyance time is extended because the conveyance length of the total of the multi-fed sheet **S1** and the multi-fed sheet **S2** becomes larger than the conveyance length of the sheet **S1**. For example, if the detection start time T_s is 30.150 and the multi-feeding occurrence time T_d is 30.220, the sheet has moved 70 msec when the multi-feeding is detected (that is, the sheet has moved by 70 mm if the speed is 1000 mm/s). The time difference is the multi-feeding shift time t that corresponds to the amount of shift between the multi-fed sheets.

[0094] Next, the multi-feeding detection control of the sheet feeding apparatus of the first embodiment will be described. If the CPU **151** proceeds to Step **S14** (see FIG. **9**) of the above-described conveyance control of the sheet feeding apparatus, the multi-feeding detection portion **SS4** receives the multi-feeding-detection start instruction from the conveyance control portion **SS2** (**S105**), so that the CPU **151** starts the detection control of the sheet feeding apparatus. In the present embodiment, the description will be made, as an example, for a case where the multi-feeding detection portion **SS4** receives the information: "sensor ID: multi-feeding detection sensor **SN12**, detection start time: 30.150, detection end time: 30.570".

[0095] As illustrated in FIG. **10**, if the CPU **151** starts the multi-feeding detection control, then the CPU **151** determines whether the detection start time T_s (i.e., 30.150) for the multi-feeding is reached (**S31**). If the detection start time T_s is not reached (**S31**: No), then the CPU **151** repeats the process until the detection start time T_s is reached. If the detection start time T_s is reached (**S31**:

Yes), then the CPU **151** starts the multi-feeding detection (S32).

[0096] Then the CPU **151** determines whether the multi-feeding has occurred (S33). The CPU **151** proceeds to Step S34 if the multi-feeding has not occurred (S33: No), or proceeds to Step S38 if the multi-feeding has occurred (S33: Yes). If the CPU **151** proceeds to Step S34, then the CPU **151** determines whether a detection end time T_e (i.e., 30.570) for the multi-feeding is reached. If the detection end time T_e is not reached (S34: No), then the CPU **151** returns to Step S33, and waits while repeating the steps S33 to S34. If the detection end time T_e is reached (S34: Yes), then the CPU **151** proceeds to Step S35. Then the CPU **151** creates information indicating that the multi-feeding has not occurred (S35), ends the multi-feeding detection (S36), and notifies the conveyance control portion SS2 of the detection result (S37). With these operations, the CPU **151** ends the multi-feeding detection control.

[0097] On the other hand, if the CPU **151** determines in the above-described Step S33 that the multi-feeding has occurred (S33: Yes), then the CPU **151** proceeds to Step S38 and obtains the multi-feeding occurrence time T_d . Then the CPU **151** creates information indicating that the multi-feeding has occurred (S39). This information contains the multi-feeding shift time t (70 msec) that corresponds to the time by which the conveyance length is extended due to the multi-feeding in which one sheet is fed together with another sheet. For example, the CPU **151** creates the information: “multi-feeding: occurred, multi-feeding shift time: 70 msec”. Then the CPU **151** ends the multi-feeding detection (S36), and notifies the conveyance control portion SS2 of the detection result (S37). With these operations, the CPU **151** ends the multi-feeding detection control.

Sheet-Conveyance-Path Switch Determination Control

[0098] Next, the sheet-conveyance-path switch determination control of the sheet feeding apparatus of the first embodiment will be described with reference to FIG. 11. FIG. 11 is a flowchart illustrating the sheet-conveyance-path switch determination control of the sheet feeding apparatus of the first embodiment.

[0099] If the CPU **151** proceeds to Step S16 (see FIG. 9) of the above-described conveyance control of the sheet feeding apparatus, then the CPU **151** starts the sheet-conveyance-path switch determination control, and determines whether the result of the multi-feeding detection indicates that the multi-feeding has occurred (S41). If the result of the multi-feeding detection indicates that the multi-feeding has not occurred (S41: No), then the CPU **151** ends the process. On the other hand, if the result of the multi-feeding detection indicates that the multi-feeding has occurred (S41: Yes), then the CPU **151** proceeds to Step S42.

[0100] Then the CPU **151** calculates a conveyance-path switch start time (S42). For example, it takes 40 msec (obtained by adding a switching-portion driving time 30 msec with a margin 10 msec) to switch the conveyance path. The time at which the leading edge of the sheet S1 reaches the error-bin branch point (P5) is 30.655 (that is, the leading edge of the sheet S1 reaches the error-bin branch point when 655 msec has elapsed since the start of conveyance of the sheet S1 from the second sheet-feeding cassette) (see FIG. 7). Thus, the conveyance-path switch start time is 30.615 obtained by subtracting 40 msec, taken for switching the conveyance path, from the time at which the leading edge of the sheet S1 reaches the error-bin branch point (P5).

[0101] Then the CPU **151** calculates a conveyance-path return start time (S43). That is, the CPU **151** calculates a time at which the trailing edge of the following sheet S2 of the multi-fed sheet S1 and the multi-fed sheet S2 passes through the error-bin branch point (P5). In the present embodiment, two examples will be described.

[0102] Example 1 is a case where the result of the multi-feeding detection indicates that the multi-feeding has occurred in the conveyance of A3 sheets, and where the multi-feeding shift time is 0 msec.

In this example, [0103] sheet length: 420 mm (A3 size), [0104] sheet conveyance speed: 1000 mm/s, and [0105] multi-feeding shift time: 0 msec.

[0106] Example 2 is a case where the result of the multi-feeding detection indicates that the multi-

feeding has occurred in the conveyance of A3 sheets, and where the multi-feeding shift time is 70 msec.

In this example, [0107] sheet length: 420 mm (A3 size), [0108] sheet conveyance speed: 1000 mm/s, and [0109] multi-feeding shift time: 70 msec.

[0110] In Example 1, the time at which the trailing edge of the following sheet S2 passes through the error-bin branch point (P5) is 31.075 obtained by adding the time 30.655 at which the leading edge of the sheets reaches the error-bin branch point (P5), with the time 420 msec taken for the trailing edge of the sheets to pass through the error-bin branch point (P5). In Example 2, the time at which the trailing edge of the following sheet S2 passes through the error-bin branch point (P5) is obtained by further adding the time at which the trailing edge of the following A3 sheet S2 passes through the error-bin branch point (P5), with the multi-feeding shift time 70 msec. Thus, the time at which the trailing edge of the following sheet S2 passes through the error-bin branch point (P5) is 31.145.

[0111] Then the CPU 151 obtains a following-sheet arrival time that serves as a third time (S44). That is, the CPU 151 obtains the time at which the sheet S3 that follows the multi-fed sheet S1 and the multi-fed sheet S2 reaches the error-bin branch point (P5), from the conveyance-route management table (see FIG. 7). That is, if the sheet S3 for the double-side printing is conveyed from the double-side-printing-path entrance EN1 (i.e., the print portion 200), the CPU 151 obtains, as the following-sheet arrival time, the conveyance time from the double-side-printing-path entrance EN1 to the error-bin branch point (P5), from the conveyance-route management table. In addition, if the sheet S3 is conveyed from the horizontal-path entrance EN2 (i.e., the sheet feeding apparatus 120), the CPU 151 obtains, as the following-sheet arrival time, the conveyance time from the horizontal-path entrance EN2 to the error-bin branch point (P5), from the conveyance-route management table.

[0112] Then the CPU 151 determines whether the following-sheet arrival time, obtained as described above, is earlier than the conveyance-path return start time calculated in Step S43 (S45). The CPU 151 proceeds to Step S46 if the following-sheet arrival time is earlier than the conveyance-path return start time (S45: Yes), or proceeds to Step S47 if the following-sheet arrival time is not earlier than the conveyance-path return start time (S45: No).

[0113] In Step S46, the CPU 151 sets 0 to the conveyance-path return start time. If the CPU 151 sets 0 to the conveyance-path return start time, the switching portion FL1 keeps the conveyance path switched to the sheet discharging port 115 side (see Step S53 of FIG. 12 described below). That is, the CPU 151 cancels the operation for returning the switching portion FL1 to the first position that is the initial position, and keeps the position (i.e., the second position) of the switching portion FL1. In this case, if the CPU 151 returns the switching portion FL1 to the first position, a jam may occur. This is because the leading edge of the following sheet S3 has already passed through the switching portion FL1 (i.e., the error-bin branch point P5) and entered the error sheet-discharging path Ps14 before the trailing edge of the sheet S2 passes through the switching portion FL1.

[0114] After that, if the CPU 151 proceeds to Step S47, then the CPU 151 creates sheet-conveyance-path switch information. In the case where the result of the multi-feeding detection indicates that the multi-feeding has occurred, and where the multi-feeding shift time is 70 msec, the sheet-conveyance-path switch information contains the data: "switch start time: 30.615, return start time: 31.145". With the completion of creating the sheet-conveyance-path switch information, the CPU 151 ends the sheet-conveyance-path switch determination control.

[0115] In the present embodiment, the control is performed under the condition in which the sheet conveyance speed at which the sheet is conveyed in the multi-feeding detection and the sheet conveyance speed at which the sheet passes through the switching portion FL1 are both 1000 mm/s. However, if the speed varies, the return start time may be calculated in consideration of the difference in speed. For example, in a case where the sheet conveyance speed at which the sheet

passes through the switching portion FL1 is 1400 mm/s, the multi-feeding shift time can be calculated as 50 msec (i.e., $70 \text{ msec} \times 1000 / 1400 = 50 \text{ msec}$).

Switching-Portion Driving Control

[0116] Next, the switching-portion driving control of the sheet feeding apparatus of the first embodiment will be described with reference to FIGS. 12, 14A, and 14B. FIG. 12 is a flowchart illustrating the switching-portion driving control of the sheet feeding apparatus of the first embodiment. FIG. 14A is a diagram illustrating a case where the switching portion can be switched after multi-fed sheets have passed through the switching portion in the sheet feeding apparatus. FIG. 14B is a diagram illustrating a case where the switching portion cannot be switched after multi-fed sheets have passed through the switching portion in the sheet feeding apparatus.

[0117] If the CPU 151 proceeds to Step S18 (see FIG. 9) of the above-described conveyance control of the sheet feeding apparatus, the conveyance-path switch control portion SS5 receives the sheet-conveyance-path switch instruction from the conveyance control portion SS2 (S109), so that the CPU 151 starts the switching-portion driving control of the sheet feeding apparatus. In the present embodiment, the description will be made for a case where the conveyance-path switch control portion SS5 receives information that, for example, contains the data: “switch start time: 30.615, return start time: 31.145”.

[0118] If the CPU 151 starts the switching-portion driving control, then the CPU 151 determines whether the switch start time (i.e., 30.615) is reached (S51). The CPU 151 repeats the process if the switch start time is not reached (S51: No), or proceeds to Step S52 if the switch start time is reached (S51: Yes).

[0119] Then the CPU 151 switches the switching portion FL1 (Flapper) from the first position that is an initial position, to the second position. That is, the CPU 151 switches the conveyance path, along which the sheet S1 and the sheet S2 are conveyed, to the error bin side (S52). Then the CPU 151 determines whether the return start time is contained in the information (S53). In a case where the return start time is not contained in the information, the conveyance-path switch control portion SS5 is notified of only the switch start time. For example, in this case, the sheet-conveyance-path switch information contains the data: “switch start time: 30.615, return start time: 0”. If the return start time is not contained in the information (S53: No), then the CPU 151 ends the switching-portion driving control. Specifically, the CPU 151 cancels the switching of the switching portion FL, and keeps the second position of the switching portion FL1. That is, the CPU 151 does not switch the conveyance path to the downstream apparatus side, on which side the sheet is discharged to the downstream apparatus.

[0120] That is, if it is determined in the above-described Step S45 (see FIG. 11) that the following-sheet arrival time is earlier than the conveyance-path return start time, and that the following sheet S3 reaches the switching portion FL1 before the trailing edge of the sheet S2 of the multi-fed sheets passes through the switching portion FL1, the return start time is not contained in the information (see S46). In this case, as illustrated in FIG. 14B, the leading edge of the following sheet S3 reaches the switching portion FL1 before the trailing edge of the sheet S2 of the fed-sheet S1 and the fed-sheet S2 passes through the switching portion FL1. If the switching portion FL1 is switched and returned to the first position at the timing at which the leading edge of the following sheet S3 reaches the switching portion FL1 before the trailing edge of the sheet S2 of the fed-sheet S1 and the fed-sheet S2 passes through the switching portion FL1, the following sheet S3 will be caught by the switching portion FL1 and a jam will occur. For this reason, the switching of the switching portion FL1 is canceled, and the switching portion FL1 is not switched. As a result, the following sheet S3 is also discharged to the sheet discharging tray 119, so that the occurrence of jam can be prevented. In the present embodiment, after the switching of the switching portion FL1 is canceled, the CPU 151 stops the conveyance rollers (see S22), and ends the conveyance control. However, the multi-fed sheet S1, the multi-fed sheet S2 and the following sheet S3 may be automatically recovered and fed.

[0121] On the other hand, if the return start time is contained in the information in the above-described Step S53 (S53: Yes), then the CPU 151 proceeds to Step S54. If the CPU 151 proceeds to Step S54, then the CPU 151 determines whether the return start time (i.e., 31.145) is reached. The CPU 151 repeats the process if the return start time is not reached (S54: No), or proceeds to Step S55 if the return start time is reached (S54: Yes).

[0122] Then the CPU 151 switches the switching portion FL1 (Flapper) from the second position to the first position that is an initial position. That is, the CPU 151 switches the conveyance path to the downstream apparatus side (S55), and ends the switching-portion driving control.

[0123] That is, in this case, it is determined that the following-sheet arrival time is not earlier than the conveyance-path return start time, and that the following sheet reaches the switching portion FL1 after the trailing edge of the sheet S2 of the multi-fed sheets passes through the switching portion FL1. Thus, as illustrated in FIG. 14A, the leading edge of the following sheet S3 reaches the switching portion FL1 after the trailing edge of the sheet S2 of the fed-sheet S1 and the fed-sheet S2 passes through the switching portion FL1. If the switching portion FL1 is switched and returned to the first position at the timing at which the leading edge of the following sheet S3 reaches the switching portion FL1 after the trailing edge of the sheet S2 of the fed-sheet S1 and the fed-sheet S2 passes through the switching portion FL1, the sheet S2 will not be caught by the switching portion FL1, causing no jam, and the following sheet S3 will also not be caught by the switching portion FL1, causing no jam. In short, in addition to making it possible to discharge the sheet S1 and the sheet S2 to the sheet discharging tray 119 and the following sheet S3 to the print portion 200 that is a downstream apparatus, the occurrence of jam can be prevented. In the present embodiment, after the switching portion FL1 is switched to the first position, the CPU 151 stops the conveyance rollers (see S22), and ends the conveyance control.

[0124] However, the multi-fed sheet S1 and the multi-fed sheet S2 may be automatically recovered and fed.

Summary of First Embodiment

[0125] As described above, if the sheet S1 and the sheet S2 are multi-fed, there is a case where the sheet S1 and the sheet S2 overlap with each other, shifted from each other. In this case, the trailing edge of the sheet S2 passes through (reaches) the switching portion FL1, delayed from the trailing edge of the sheet S1 by the amount of shift. However, in the sheet feeding apparatus 110, the control portion 150 (i.e., the CPU 151) calculates the conveyance-path return start time at which the trailing edge of the sheet S2 passes through the switching portion FL1, in a case where the multi-feeding detection sensor detects the multi-feeding. In addition, the control portion 150 switches the switching portion FL1 from the second position to the first position at the conveyance-path return start time. With this operation, the multi-fed sheet S1 and the multi-fed sheet S2 can be appropriately guided to the error sheet-discharging path Ps14, and the switching portion FL1 can be switched after the trailing edge of the sheet S2 passes through the switching portion FL1. As a result, the occurrence of jam can be prevented.

[0126] In addition, in a case where the multi-feeding detection sensor detects the multi-feeding, the control portion 150 switches the switching portion FL1 from the second position to the first position in accordance with the amount of shift (i.e., the multi-feeding shift time t) between the leading edge of the sheet S1 and the leading edge of the sheet S2 in the conveyance direction V. With this operation, the control portion 150 can switch the switching portion FL1 after the trailing edge of the sheet S2 passes through the switching portion FL1.

[0127] In addition, the control portion 150 calculates the conveyance-path return start time by adding the second time at which the trailing edge of the sheet S1 passes through the switching portion FL1, with the length of time from the time when the multi-feeding detection sensor detects the leading edge of the sheet S1, to the time when the multi-feeding detection sensor detects the multi-feeding. That is, the control portion 150 calculates the conveyance-path return start time by adding the detection end time T_e with a time (i.e., the multi-feeding shift time t) from the detection

start time T_s to the multi-feeding occurrence time T_d illustrated in FIG. 13. With this operation, the control portion **150** can calculate the time at which the trailing edge of the sheet **S2** passes through the switching portion **FL1**, as the conveyance-path return start time.

[0128] In addition, in a case where the sheet **S3** is conveyed following the sheet **S2**, and where the following-sheet arrival time at which the leading edge of the sheet **S3** passes through the switching portion **FL1** is earlier than the conveyance-path return start time, the control portion **150** cancels the switching of the switching portion **FL1** to be performed at the conveyance-path return start time. With this operation, it is prevented that the switching portion **FL1** is switched after the leading edge of the sheet **S3** passes through the switching portion **FL1**. As a result, the occurrence of jam caused by the sheet **S3** being caught by the switching portion **FL1** can be prevented.

[0129] In addition, in a case where the switching portion **FL1** is positioned in the first position in the sheet feeding apparatus **110**, the sheet can be guided to the third horizontal-conveyance path **Ps13**, along which the sheet is conveyed to the ink-jet head **201** that forms an image. If multi-fed sheets are conveyed to the ink-jet head **201** that forms an image by using the ink-jet system, the ink-jet head **201** may be damaged by the multi-fed sheets. In the sheet feeding apparatus **110**, however, the switching portion **FL1** is switched appropriately, so that the multi-fed sheets are not conveyed to the ink-jet head **201**. As a result, the damage of the ink-jet head **201** can be prevented. In particular, since the sheet feeding apparatus **110** can cause the switching portion **FL1** to guide and discharge the multi-fed sheets to the sheet discharging tray **119**, the error sheets can be removed.

Second Embodiment

[0130] Next, a second embodiment will be described with reference to FIGS. 15 and 16. In the second embodiment, part of the first embodiment is changed. FIG. 15 is a schematic cross-sectional view illustrating a sheet feeding apparatus of the second embodiment. FIG. 16 is a flowchart illustrating the conveyance control of the sheet feeding apparatus of the second embodiment. Note that in the description of the second embodiment, a component identical to a component of the above-described first embodiment is given an identical symbol, and the description thereof will be omitted.

[0131] In the above-described first embodiment, the description has been made for the case where the time at which the trailing edge of the sheet **S2** of the multi-fed sheets passes through the switching portion **FL1** is calculated. In the second embodiment, however, a conveyance sensor **SN4** is disposed on the second horizontal-conveyance path **Ps12**, as a trailing-edge detection portion that detects the trailing edge of the sheet **S2**.

[0132] Specifically, as illustrated in FIG. 15, the sheet feeding apparatus **110** of the second embodiment includes the conveyance sensor **SN4** that conveys the sheet that passes through the second horizontal-conveyance path **Ps12**. That is, the conveyance sensor **SN4** is disposed upstream of the switching portion **FL1** (i.e., the error-bin branch point **P5**) in the conveyance direction **V**, and downstream of the horizontal-path meeting point **P4** in the conveyance direction **V**. The conveyance sensor **SN4** detects the trailing edge of the sheet **S2** when the multi-fed sheet **S1** and the multi-fed sheet **S2** passes through the conveyance sensor **SN4**. Preferably, the conveyance sensor **SN4** is disposed upstream of the switching portion **FL1**, spaced from the switching portion **FL1** by a predetermined distance in consideration of the response time from the time when the trailing edge of the sheet **S2** is detected by the conveyance sensor **SN4**, to the time when the switching portion **FL1** is driven.

Control of Sheet Feeding Apparatus of Second Embodiment

[0133] Next, control of the sheet feeding apparatus of the second embodiment, in particular, conveyance control in the second embodiment will be described with reference to FIG. 16. Note that the conveyance control of the second embodiment differs from the conveyance control of the first embodiment (see FIG. 9) in the steps **S61** to **S63** and **S71** to **S72**, and the other steps of the conveyance control of the second embodiment are substantially the same as those of the

conveyance control of the first embodiment.

[0134] As in the above-described first embodiment, if the conveyance control portion SS2 receives, in Step S101, an instruction for feeding and conveying a sheet, from the sheet-feeding-apparatus management portion SS1, the conveyance control of the sheet feeding apparatus is started. Then, as illustrated in FIG. 16, the CPU 151 drives the conveyance roller (S11). Then the CPU 151 determines whether the instruction for feeding and conveying a sheet is an instruction for feeding a sheet from any one of the first to third sheet-feeding trays 111 to 113, or an instruction for receiving the sheet S3 conveyed from an upstream apparatus (i.e., another sheet feeding apparatus or the print portion 200) disposed upstream in the conveyance direction V (S12). The CPU 151 proceeds to Step S13 if the instruction for feeding and conveying a sheet is an instruction for feeding a sheet from any one of the first to third sheet-feeding trays 111 to 113, or proceeds to Step S23 if the instruction for feeding and conveying a sheet is an instruction for receiving the sheet S3 conveyed from the upstream apparatus.

[0135] If the CPU 151 proceeds to Step S13, then the CPU 151 notifies the sheet-feeding control portion SS3 of a sheet-feeding start instruction. When the CPU 151 notifies the sheet-feeding control portion SS3 of the sheet-feeding start instruction, the CPU 151 notifies the sheet-feeding control portion SS3 (i.e., the sheet-feeding separation portion) of information that, for example, contains the data: “cassette ID: second sheet-feeding cassette, sheet-conveyance start time: 12:10:30 and 000 milliseconds” (the sheet-conveyance start time, 12:10:30 and 000 milliseconds, is hereinafter abbreviated as 30.000). Upon receiving the sheet-feeding start instruction, the sheet-feeding control portion SS3 performs a process for feeding a sheet, and starts to feed the sheet from the second sheet-feeding tray 112 at the sheet-conveyance start time of 30.000.

[0136] Then the CPU 151 notifies the multi-feeding detection portion SS4 (i.e., the multi-feeding detection sensor) of a multi-feeding-detection start instruction (S14). When the CPU 151 notifies the multi-feeding detection portion SS4 of the multi-feeding-detection start instruction, the CPU 151 notifies the multi-feeding detection portion SS4 of information that, for example, contains the data: “sensor ID: multi-feeding detection sensor SN12, detection start time: 30.150, detection end time: 30.570” if an A3-size sheet (whose length is 420 mm) is conveyed at a speed of 1000 mm/s. The detection start time is determined as described above because the leading edge of the fed sheet passes the multi-feeding detection sensor SN12 when 150 msec has elapsed since the sheet conveyance time. The detection end time is determined, based on a time of 420 msec in which the sheet having a length of 420 mm passes through the multi-feeding detection sensor SN12 (the time of period in which the detection is performed may be determined based on the sheet length, or may be determined based on another parameter). Upon receiving the multi-feeding-detection start instruction, the multi-feeding detection portion SS4 performs a process for detecting the multi-feeding, and starts the detection of the multi-feeding at the detection start time of 30.150.

[0137] After that, in the second embodiment, the CPU 151 determines whether the conveyance sensor SN4 has detected the leading edge of the sheet S1 (S61). If the conveyance sensor SN4 has not detected the leading edge of the sheet S1 (S61: No), then the CPU 151 repeats the process until the conveyance sensor SN4 detects the leading edge of the sheet S1. If the conveyance sensor SN4 has detected the leading edge of the sheet S1 (S61: Yes), then the CPU 151 proceeds to Step S62.

[0138] If the CPU 151 proceeds to Step S62, then the CPU 151 determines whether the result of the multi-feeding detection, sent from the multi-feeding detection portion SS4, indicates that the multi-feeding has occurred (S62). The CPU 151 proceeds to Step S63 if the CPU 151 determines that the multi-feeding has occurred (S62: Yes), or proceeds to Step S27 if the CPU 151 determines that the multi-feeding has not occurred (S62: No). In a case where the multi-feeding has occurred and the CPU 151 proceeds to Step S63, the leading edge of the multi-fed sheet S1 has been detected in the above-described Step S61, by the conveyance sensor SN4. In this case, the CPU 151 then sends an instruction for switching the switching portion FL1 (Flapper). That is, the CPU 151 drives the switching portion FL1 and switches the switching portion FL1 from the first position that is an

initial position, to the second position if the conveyance sensor SN4 has detected the sheet S1 and the multi-feeding has occurred.

[0139] Then the CPU 151 notifies the sheet-feeding-apparatus management portion SS1 of the occurrence of multi-feeding (S19). In addition, the occurrence of multi-feeding is notified to the print portion 200 via the sheet-feeding-apparatus management portion SS1. The notification may be performed when the sheet S1 and the sheet S2 are discharged to the sheet discharging port 115. However, if the notification is performed earlier, the following sheet S3 can be stopped earlier, and the operation of the print portion 200 can be stopped or changed earlier.

[0140] Then the CPU 151 determines whether the conveyance sensor SN4 has detected the trailing edge of the sheet S2 of the multi-fed sheets (S71). If the conveyance sensor SN4 has not detected the trailing edge of the sheet S2 (S71: No), then the CPU 151 repeats the process until the conveyance sensor SN4 detects the trailing edge of the sheet S2. If the conveyance sensor SN4 has detected the trailing edge of the sheet S2 (S71: Yes), then the CPU 151 proceeds to Step S72.

[0141] If the CPU 151 proceeds to Step S72, then the CPU 151 sends an instruction for returning the switching portion FL1 (Flapper) to the first position. With this operation, the switching portion FL1 can be returned from the second position to the first position when the trailing edge of the sheet S2 passes through the switching portion FL1. This is because the trailing edge of the sheet S2 which has passed through the switching portion FL1 in a state where the sheet S2 is shifted from the sheet S1 due to the multi-feeding has been detected by the conveyance sensor SN4. As a result, the sheet S2 can be prevented from being caught by the switching portion FL1, so that the occurrence of jam can be prevented.

[0142] Then the CPU 151 determines whether the sheet S1 and the sheet S2 are discharged to the sheet discharging port 115 (S20). The completion of discharging sheets may be determined based on time. In another case, a sensor for monitoring the conveyance of sheets may be disposed in the vicinity of the sheet discharging port 115, and the sensor may monitor the passage of the trailing edge of the sheet S2. The sheet-discharging completion time can be calculated by adding the time at which the leading edge of the sheet S1 reaches the sheet discharging port 115 (when 905 msec has elapsed since the start of conveyance of the sheet S1 from the second sheet-feeding cassette) (see FIG. 7), with a time that corresponds to a length of a multi-fed sheet (including the amount of shift in the multi-feeding). If the sheet discharging is not completed (S20: No), then the CPU 151 repeats the process until the sheet discharging is completed. If the sheet discharging is completed (S20: Yes), then the CPU 151 proceeds to Step S21.

[0143] If the CPU 151 proceeds to Step S21, then the CPU 151 notifies the conveyance control portion SS2 of the completion of discharging the sheets to the error bin (i.e., the sheet discharging port 115). With this operation, the CPU 151 stops the sheet discharging roller (S22), and ends the conveyance control of the sheet feeding apparatus 110.

[0144] On the other hand, as described above, if the instruction for feeding and conveying a sheet is, in Step S12, an instruction for receiving the sheet S3 conveyed from the above-described upstream apparatus, then the CPU 151 proceeds to Step S23. If the CPU 151 proceeds to Step S23, then the CPU 151 determines whether a sheet reception time is reached. If the sheet reception time is not reached (S23: No), then the CPU 151 repeats the process until the sheet reception time is reached. If the sheet reception time is reached (S23: Yes), then the CPU 151 proceeds to Step S24. If the CPU 151 proceeds to Step S24, then the CPU 151 receives the sheet S3 from the upstream apparatus.

[0145] Then the CPU 151 determines whether a passage time at which the sheet S3 passes through the error-bin branch point (P5) is reached (S25). If the passage time is not reached (S25: No), then the CPU 151 repeats the process until the passage time is reached. If the passage time is reached (S25: Yes), then the CPU 501 proceeds to Step S26.

[0146] In Step S26, the CPU 151 determines whether the direction of the conveyance path extends toward a downstream apparatus disposed downstream in the conveyance direction V (i.e., the print

portion **200**, the sheet feeding apparatus **110** if the sheet is fed from the sheet feeding apparatus **120**, or the sheet feeding apparatus **120** if the sheet is fed from the sheet feeding apparatus **130**). That is, the CPU **151** determines whether the switching portion **FL1** is positioned in the initial position. If the direction of the conveyance path extends toward the downstream apparatus (**S26**: Yes), then the CPU **151** proceeds to Step **S27**.

[0147] If the CPU **151** proceeds to Step **S27** in a case where the multi-feeding has not occurred in the above-described Step **S62** (**S62**: No), or where the direction of the conveyance path extends toward the downstream apparatus (**S26**: Yes), then the CPU **151** determines whether the sheet **S1** or **S3** has been discharged to the downstream apparatus. The completion of discharging the sheet to the downstream apparatus may be determined based on time. In another case, a sensor for monitoring the conveyance of sheets may be disposed in the vicinity of the sheet discharging port **EX1**, and the sensor may monitor the passage of the trailing edge of the sheet **S1** or **S3**. The sheet-discharging completion time can be calculated by adding the time at which the leading edge of the sheet reaches the sheet discharging port **EX1** (when 705 msec has elapsed since the start of conveyance of the sheet **S1** from the second sheet-feeding cassette) (see FIG. 7), with a time that corresponds to a length of the sheet. If the sheet discharging is not completed (**S27**: No), then the CPU **151** repeats the process until the sheet discharging is completed. If the sheet discharging is completed (**S27**: Yes), then the CPU **151** proceeds to Step **S28**. If the CPU **151** proceeds to Step **S28**, then the CPU **151** notifies the conveyance control portion **SS2** of the completion of normally delivering the sheet **S1** or **S3** to the downstream apparatus, and proceeds to Step **S22**. With this operation, the CPU **151** stops the sheet discharging roller (**S22**), and ends the conveyance control of the sheet feeding apparatus **110**.

[0148] On the other hand, if the direction of the conveyance path does not extend toward the downstream apparatus in the above-described Step **S26** (**S26**: No), that is, if the switching portion **FL1** has been switched and the direction of the conveyance path extends toward the sheet discharging port **115**, then the CPU **151** proceeds to the above-described Step **S71**. In this case, the preceding sheet **S1** and the sheet **S2** have been multi-fed, and the sheet **S3** conveyed from the upstream apparatus has reached the switching portion **FL1** before the switching portion **FL1** is returned to the first position, because the amount of shift is large. Thus, in this case, the switching portion **FL1** cannot be returned to the first position.

[0149] If the CPU **151** proceeds to Step **S71** in this manner, then the CPU **151** determines whether the conveyance sensor **SN4** has detected the trailing edge of the sheet **S3** that follows the multi-fed sheet **S1** and the multi-fed sheet **S2** (**S71**). If the conveyance sensor **SN4** has not detected the trailing edge of the sheet **S3** (**S71**: No), then the CPU **151** repeats the process until the conveyance sensor **SN4** detects the trailing edge of the sheet **S3**. If the conveyance sensor **SN4** has detected the trailing edge of the sheet **S3** (**S71**: Yes), then the CPU **151** proceeds to Step **S72**.

[0150] If the CPU **151** proceeds to Step **S72**, then the CPU **151** sends an instruction for returning the switching portion **FL1** to the first position. With this operation, the switching portion **FL1** can be returned from the second position to the first position when the trailing edge of the sheet **S3** passes through the switching portion **FL1**. This is because the trailing edge of the sheet **S3** following the multi-fed sheet **S1** and the multi-fed sheet **S2** and received from the upstream apparatus has been detected by the conveyance sensor **SN4**, in addition to the detection of the multi-fed sheet **S1** and the multi-fed sheet **S2**. As a result, the sheet **S3** can be prevented from being caught by the switching portion **FL1**, so that the occurrence of jam can be prevented.

[0151] Then the CPU **151** determines whether the sheets **S1** to **S3** are discharged to the sheet discharging port **115** (**S20**). If the sheet discharging is not completed (**S20**: No), then the CPU **151** repeats the process until the sheet discharging is completed. If the sheet discharging is completed (**S20**: Yes), then the CPU **151** notifies the conveyance control portion **SS2** of the completion of discharging the sheets to the error bin (i.e., the sheet discharging port **115**). With this operation, the CPU **151** stops the sheet discharging roller (**S22**), and ends the conveyance control of the sheet

feeding apparatus **110**.

[0152] Note that in the above-described conveyance control of the sheet feeding apparatus **110**, the plurality of conveyance rollers that serve as the conveyance portion **110A** is collectively driven or stopped. However, only necessary conveyance rollers may be driven or stopped in a sequential manner by calculating the time at which the sheet reaches a corresponding component.

[0153] In addition, although the leading edge of a multi-fed sheet is detected by the conveyance sensor **SN4** in the present embodiment, the passage of the leading edge of the multi-fed sheet may be determined, based on time. This is because the passage time of the leading edge of a sheet in a case where the sheet is multi-fed is the same as the passage time of the leading edge of the sheet in a case where the sheet is not multi-fed.

Summary of Second Embodiment

[0154] As described above, if the sheet **S1** and the sheet **S2** are multi-fed, there is a case where the sheet **S1** and the sheet **S2** overlap with each other, shifted from each other. In this case, the trailing edge of the sheet **S2** passes through (reaches) the switching portion **FL1**, delayed from the trailing edge of the sheet **S1** by the amount of shift. However, in the sheet feeding apparatus **110**, in a case where the control portion **150** (i.e., the CPU **151**) detects the multi-feeding by using the multi-feeding detection sensor, the conveyance sensor **SN4** detects that the trailing edge of the sheet **S2** has passed through the switching portion **FL1**. In addition, if the conveyance sensor **SN4** detects that the trailing edge of the sheet **S2** has passed through the switching portion **FL1**, the control portion **150** switches the switching portion **FL1** from the second position to the first position. With this operation, the multi-fed sheet **S1** and the multi-fed sheet **S2** can be appropriately guided to the error sheet-discharging path **Ps14**, and the switching portion **FL1** can be switched after the trailing edge of the sheet **S2** passes through the switching portion **FL1**. As a result, the occurrence of jam can be prevented.

[0155] In addition, in a case where the sheet **S3** is conveyed following the sheet **S2**, if the timing at which the leading edge of the sheet **S3** passes through the switching portion **FL1** is earlier than the timing at which the trailing edge of the sheet **S2** passes through the switching portion **FL1**, the control portion **150** switches the switching portion **FL1** to the first position after the conveyance sensor **SN4** detects that the trailing edge of the sheet **S3** has passed through the switching portion **FL1**. With this operation, the switching portion **FL1** is switched after the trailing edge of the sheet **S3** passes through the switching portion **FL1**. As a result, the occurrence of jam caused by the sheet **S3** being caught by the switching portion **FL1** can be prevented.

[0156] Since the other configuration, operations, and effects of the second embodiment are the same as those of the above-described first embodiment, the description thereof will be omitted.

[0157] The present invention can prevent the occurrence of jam while guiding the multi-fed first and second sheets to the third conveyance path.

Other Embodiments

[0158] In the above-described first and second embodiments, in a case where the multi-feeding is detected in the sheet feeding apparatus **110**, multi-fed sheets are discharged to the sheet discharging tray **119**. However, the present disclosure is not limited to this configuration. That is, the present disclosure is applicable to any configuration as long as multi-fed sheets are conveyed, in the configuration, to a portion other than the portion to which normal sheets are conveyed. For example, the present invention may be applied to a portion of the fixing portion **300** in which error sheets are discharged toward the sheet discharging port **301**. Similarly, the present invention may be applied to a portion of the reversing portion **500** in which error sheets are discharged toward the sheet discharging port **501**. In a case where the present invention is applied to the reversing portion **500**, the destination to which normal sheets (which are not unacceptable) are conveyed is a discharging tray for the normal sheets. That is, the destination to which normal sheets are conveyed may not be the image forming portion.

[0159] In addition, in the first and second embodiments, the description has been made for the case

where time is used for managing the position of a sheet. However, the present disclosure is not limited to this. For example, the position of a sheet may be managed by using the time that is counted from a reference time (e.g., the time counted from when power is turned on, or the time counted from a time specified for starting to feed a sheet). In another case, the position of a sheet may be managed by using the coordinates at which the sheet is located in the apparatus, and the time in which the sheet has moved may be calculated by using the coordinates and the conveyance speed.

[0160] In addition, in the first embodiment, the description has been made for the case where the time (i.e., the conveyance-path return start time) at which the trailing edge of the sheet S2 passes through the switching portion FL1 is calculated in accordance with the amount of shift (i.e., the multi-feeding shift time t) (see FIG. 13) between the multi-fed sheet S1 and the multi-fed sheet S2. However, the time at which the trailing edge of the sheet S2 passes through the switching portion FL1 may be calculated by using any method. For example, the leading edge of the sheet S2 (i.e., the multi-feeding occurrence time T_d) may be detected, and then the multi-feeding occurrence time T_d may be added with the time obtained by dividing the length of the sheet S2 in the conveyance direction V by the conveyance speed.

[0161] In addition, since the image forming system 1 described in the first and second embodiments is one example, the system to which the present disclosure is applied may be any system as long as the system has a configuration for conveying sheets and causing the switching portion to switch the conveyance path. In particular, the system to which the present disclosure is applied is not limited to a system in which the image forming portion has an ink-jet system. For example, the system to which the present disclosure is applied may be a system in which the image forming portion has an electrophotographic system.

[0162] The present disclosure may be embodied by providing a program, which achieves one or more functions of the above-described embodiments, to a system or an apparatus via a network or a storage medium, and by causing one or more processors of the system or the apparatus to read and execute the program. In addition, the present disclosure may be embodied by a circuit (e.g., an ASIC) that achieves one or more functions.

[0163] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0164] This application claims the benefit of Japanese Patent Application No.2024-035939, filed Mar. 8, 2024, which is hereby incorporated by reference herein in its entirety.

Claims

1. A sheet conveyance apparatus comprising: a conveyance portion configured to convey a sheet; a switching portion configured to be switched between a first position and a second position, the first position being a position at which the switching portion guides the sheet conveyed in a first conveyance path by the conveyance portion, to a second conveyance path, the second position being a position at which the switching portion guides the sheet conveyed in the first conveyance path by the conveyance portion, to a third conveyance path; a multi-feeding detection portion disposed upstream of the switching portion in a sheet conveyance direction and configured to detect multi-feeding in which a first sheet and a second sheet that follows the first sheet overlap with each other; and a control portion configured to control the switching portion for guiding the first sheet and the second sheet to the third conveyance path by switching the switching portion to the second position in a case where multi-feeding is detected by the multi-feeding detection portion, wherein in the case where multi-feeding is detected by the multi-feeding detection portion, the control portion is configured to calculate a first time at which a trailing edge of the second sheet

in the sheet conveyance direction has been passed through the switching portion, and switch the switching portion from the second position to the first position at the first time.

2. The sheet conveyance apparatus according to claim 1, wherein in the case where multi-feeding is detected by the multi-feeding detection portion, the control portion is configured to switch the switching portion from the second position to the first position based on an amount of shift between a leading edge of the first sheet and a leading edge of the second sheet in the sheet conveyance direction.

3. The sheet conveyance apparatus according to claim 1, wherein the control portion is configured to calculate the first time by adding a length of time from a time when the multi-feeding detection portion detects a leading edge of the first sheet to a time when the multi-feeding detection portion detects multi-feeding, to a second time at which the trailing edge of the first sheet in the sheet conveyance direction passes through the switching portion.

4. The sheet conveyance apparatus according to claim 1, wherein in a case where a third sheet that follows the second sheet is conveyed, if a third time at which a leading edge of the third sheet in the sheet conveyance direction passes through the switching portion is earlier than the first time, the control portion is configured to cancel switching of the switching portion to be performed at the first time.

5. The sheet conveyance apparatus according to claim 1, wherein the second conveyance path is a conveyance path for conveying the sheet to an image forming portion configured to form an image on the sheet.

6. The sheet conveyance apparatus according to claim 5, wherein the image forming portion is configured to form an image by using an ink-jet system.

7. The sheet conveyance apparatus according to claim 1, further comprising: a discharging support portion configured to support the sheet discharged, wherein the third conveyance path is a conveyance path for conveying the sheet to the discharging support portion.

8. The sheet conveyance apparatus according to claim 1, further comprising: a support portion configured to support a plurality of sheets; and a feeding portion configured to feed the sheets supported by the support portion to the first conveyance path.

9. A sheet conveyance apparatus comprising: a conveyance portion configured to convey a sheet; a switching portion configured to be switched between a first position and a second position, the first position being a position at which the switching portion guides the sheet conveyed in a first conveyance path by the conveyance portion, to a second conveyance path, the second position being a position at which the switching portion guides the sheet conveyed in the first conveyance path by the conveyance portion, to a third conveyance path; a multi-feeding detection portion disposed upstream of the switching portion in a sheet conveyance direction and configured to detect multi-feeding in which a first sheet and a second sheet that follows the first sheet overlap with each other; a trailing-edge detection portion disposed upstream of the switching portion in the sheet conveyance direction and configured to detect a trailing edge of the second sheet in the sheet conveyance direction; and a control portion configured to control the switching portion for guiding the first sheet and the second sheet to the third conveyance path by switching the switching portion to the second position in a case where multi-feeding is detected by the multi-feeding detection portion, wherein in the case where multi-feeding is detected by the multi-feeding detection portion, the control portion is configured to switch the switching portion from the second position to the first position based on detecting the trailing edge of the second sheet by the trailing-edge detection portion.

10. The sheet conveyance apparatus according to claim 9, wherein the trailing-edge detection portion is disposed with a distance from the switching portion in the sheet conveyance direction.

11. The sheet conveyance apparatus according to claim 9, wherein the control portion is configured to switch the switching portion to the second position in a case where the trailing-edge detection portion detects a leading edge of the first sheet in the sheet conveyance direction and the multi-

feeding detection portion detects multi-feeding.

12. The sheet conveyance apparatus according to claim 9, wherein the second conveyance path is a conveyance path for conveying the sheet to an image forming portion configured to form an image in the sheet.

13. The sheet conveyance apparatus according to claim 12, wherein the image forming portion is configured to form an image by using an ink-jet system.

14. The sheet conveyance apparatus according to claim 9, further comprising: a discharging support portion configured to support the sheet discharged, wherein the third conveyance path is a conveyance path for conveying the sheet to the discharging support portion.

15. The sheet conveyance apparatus according to claim 9, further comprising: a support portion configured to support a plurality of sheets; and a feeding portion configured to feed the sheets supported by the support portion to the first conveyance path.

16. An image forming system comprising: the sheet conveyance apparatus according to claim 1; and an image forming apparatus configured to form an image on the sheet conveyed from the sheet conveyance apparatus.
